TeachMyPC

Business Requirements Document | Version 1.0 | 16 October 2026
macOS-first hackathon MVP | Product owner: TeachMyPC team

Product promise: Teach your computer once. Let it handle the repetitive work.

1. Executive Summary

TeachMyPC is a privacy-first macOS desktop utility that turns natural-language instructions into user-approved, local file automation rules. The first release allows a user to describe a repetitive file task, inspect a structured proposal generated by Gemma 3 through local Ollama, select permitted folders, preview the result, and activate a deterministic automation. The model interprets intent; it never executes filesystem actions.

2. Problem & Opportunity

Everyday computer users repeatedly rename, copy, and organize files. Existing automation products can be powerful but often expose complex concepts such as triggers, conditions, scripts, and permissions. TeachMyPC aims to make one narrow, useful workflow understandable without programming knowledge. This is a product hypothesis, not a validated claim of market uniqueness.

3. Target Users and Jobs To Be Done

- Primary: non-technical macOS users who routinely organize downloaded files, screenshots, and documents.
- Secondary: students, office workers, freelancers, creators, and developers.
- Core job: “When a new file of a certain type appears in a folder I choose, perform a safe, predictable file action automatically.”

4. Goals and Success Criteria

ID	Goal	Acceptance evidence
BO-01	Create rules without code	User creates a rule from plain English without writing syntax.
BO-02	Meaningful open-source AI	Local Gemma proposes a structured rule that passes validation.
BO-03	Safety and trust	No activation without preview, selected folders, and approval.
BO-04	Real utility	A newly added test file triggers a real file operation.
BO-05	Privacy	No cloud AI requests; no file-content upload.
BO-06	User validation	One supervised user trial and documented feedback, if time allows.

5. MVP Scope

In scope (P0)

- macOS desktop app using Angular and Tauri 2; local Ollama running Gemma 3 4B.
- Create rule from natural-language input; clear loading, invalid-model, and unavailable-Ollama states.
- Strict parsing and schema validation; unsupported or ambiguous requests fail safely.
- Human-readable editable preview, with explicit user approval before activation.
- Native folder selection for authorized source and destination directories.
- One trigger: new file created in a selected, non-recursively watched folder.
- File-extension filter and three supported actions: rename, move, copy.
- Safe dry-run preview using a sample filename; collision handling without overwrite.
- Save, list, enable, disable, and delete rules; local activity log and error states.

Out of scope (initial release)

- System-wide app control, accessibility automation, keyboard/mouse recording, shell execution, arbitrary scripts.
- Cloud accounts, sync, payment, marketplace, plugins, Windows distribution.
- Recursive folder watching, file-content interpretation, cross-volume move, destructive deletion.
- Guaranteed Arabic parsing, auto-start at login, general undo across arbitrary apps.

6. Functional Requirements

ID	Requirement	Priority
FR-01	User enters a plain-language instruction	P0
FR-02	Local AI generates a proposed rule	P0
FR-03	App validates schema and rejects unsupported output	P0
FR-04	Preview displays trigger, filter, actions, and missing inputs	P0
FR-05	User explicitly chooses and authorizes directories	P0
FR-06	User approves and activates a validated rule	P0
FR-07	User can manage saved rules and enabled state	P0
FR-08	Watcher detects stable new files and deduplicates events	P0
FR-09	Executor performs rename, copy, or safe move	P0
FR-10	Activity log records success, skipped, and failure	P0
FR-11	Dry-run previews effect without filesystem changes	P0
FR-12	Arabic instructions and optional undo	P1

7. User Journeys

A. Create and approve

User types: “Whenever a PDF is added to my Downloads folder, move it to my PDFs folder.” The model proposes a file-created trigger, PDF filter, and move action. The app requests the actual source and destination directories using a native picker, shows a sample preview, and enables the rule only after the user approves it.

B. Run automatically

A PDF appears in the selected source directory. The watcher waits until the file is stable, checks the extension and enabled rule, validates permissions and collisions, executes the move when safe, and logs the outcome.

C. Failure and recovery

If the destination is unavailable, the file is a symlink, a filename collides, or a rule becomes invalid, no overwrite occurs. The operation is skipped or failed with a visible explanation. The source file is preserved.

8. Business Rules & Safety Requirements

- Model output is untrusted data. Only allowlisted trigger/action enums and validated templates may be accepted.
- The model must not assign executable absolute paths or folder authorization identifiers. Directory IDs come from user selection.
- Activation requires explicit consent; saving a draft is not equivalent to enabling it.
- Do not follow symlinks, silently overwrite, or process output folders as new inputs.
- Avoid duplicate execution from repeated watcher events; require file stability before processing.
- For the MVP, reject unsupported cross-volume moves rather than implementing risky copy-then-delete behavior.
- Use conservative, explicit error messages; never claim a rule succeeded before the filesystem confirms it.

9. Non-Functional Requirements

Category	Requirement
Privacy	Ollama runs locally; no third-party AI service or telemetry by default.
Security	Least-privilege folder access, strict IPC surface, no arbitrary shell execution.
Reliability	Deterministic operations; safe collisions; idempotency against repeated events.
Usability	Understandable create-preview-approve-run journey; accessible error feedback.
Performance	Responsive UI while model runs; timeouts and clear connection failures.
Maintainability	Modular boundaries, typed contracts, unit tests, and documented tradeoffs.

10. Dependencies, Constraints & Assumptions

- A macOS machine with compatible Tauri 2 build dependencies and local Ollama installed.
- Gemma 3 4B must be pulled and available; model speed depends on available memory and hardware.
- User grants access to chosen folders; OS permissions can restrict some locations.
- Hackathon submission timeline is short; prioritize one complete end-to-end workflow.
- User-generated natural language may be ambiguous; clarification or rejection is preferable to guessing.

11. Risks & Mitigations

Risk	Mitigation
AI invents unsupported actions	Strict schema, enum validation, user-facing rejection.
Accidental file loss	No overwrite, source preservation, dry-run, test fixtures.
Watcher emits repeated events	Debounce, stability check, per-file operation deduplication.
Tauri/macOS permission complexity	Choose folders explicitly and keep watcher scope narrow.
Deadline pressure	Ship P0 vertical slice; cut P1 and nonessential polish first.
Existing automation competitors	Validate a simpler first-use experience and demonstrate it honestly.

12. Monetization Hypothesis

Potential freemium model: free tier with a small number of active rules; optional one-time Pro license for advanced conditions, reusable templates, and unlimited rules. An indicative $9-$19 price range is only a hypothesis. Payment processing and license enforcement are excluded from the hackathon MVP.

13. Demo & Acceptance Checklist

- Start Ollama and open TeachMyPC.
- Enter one plain-language PDF or screenshot automation request.
- Show structured AI interpretation and explain that no actions have run.
- Choose source and destination folders; show dry-run and approve.
- Drop a test file into the source folder; show the real outcome.
- Show activity log and one safe failure case, such as a filename collision.
- Show an unsupported request being rejected instead of executed.
- Document one limitation and one realistic future improvement.

14. Release Criteria

Release the hackathon MVP only when the end-to-end happy path works on macOS, invalid AI output cannot execute, files are not overwritten, core domain tests pass, and a short reproducible demo can be recorded. The product does not need general-purpose OS automation to meet this definition.

15. Future Roadmap (Not Committed)

Arabic support improvements; richer conditions; reversible file operations; curated rule templates; app-aware triggers; global shortcuts; Windows support; plugin ecosystem. Each requires separate validation and platform-specific security review.